Part 4 – Raspberry

Component	Function
SoS	Processes instructions and controls the system.
GPIO	Connects external electronic components.
USB	Connects external devices.
HDMI	Outputs video and audio to a display.
Power	Supplies electricity to the device.
microSD	Stores the operating system and files.
Network	Provides wired internet connection.
Wireless	Provides Wi-Fi and Bluetooth connectivity.

Reflection

Q) Why are many components integrated into a single board in embedded computers such as the Raspberry Pi?

Many components are integrated into a single board in embedded computers such as the Raspberry Pi to make the device smaller, cheaper, and more efficient. By combining all components like the processor, memory, storage, and networking on one board reduces the need for extra hardware and cables. This design also lowers power consumption and improves performance because the parts can communicate faster with each other. A single-board system is easier to use, transport, and maintain, making it ideal for education, robotics, automation, and Internet of Things (IoT) projects. Integration also helps manufacturers reduce production costs and increase reliability by minimizing separate parts and connections